Direct & Inverse Proportion

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Direct proportion

What is direct proportion?

- **Direct** proportion
  - As one quantity **increases/decreases** by a certain rate (factor)
  - The other quantity will **increase/decrease** by the same rate
- The **ratio** of the two quantities is **constant**
  - E.g. 2 boxes of cereal is 800 g of cornflakes
  - **Doubling** the number of boxes of cereal (4 boxes) will **double** the amount of cornflakes (1600 g)

How do I solve direct proportion questions?

- **Read** through wordy **direct proportion** questions **carefully**
  - Ensure that you understand the **context** of the question
  - Some questions may tell you the relationship between the two values as a **ratio**
- **Identify** the two quantities involved
  - E.g. Hours worked and pay
- **Find** the **factor** that you will be **increasing/decreasing** by
  - This may be given to you in the question, e.g. 'the amount is tripled'
    - The quantity is multiplied by three
  - Alternatively, find the factor by dividing the 'new' quantity by the 'old' quantity
- **Multiply** the other quantity by this factor to find the required **quantity**
  - E.g. If three times as many hours are worked, the pay will be three times more in total
- Give your **final answer** in **context**
  - **Round** and **give units** where appropriate

How do I use the unitary method with direct proportion?

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- For example, find the weight of 7 boxes, if 8 boxes weigh 60 kg
  - Find the weight of **1 box** (1 **unit**) using division
    - 60 kg ÷ 8 boxes = 7.5 kg per box
  - Scale this unit up using multiplication
    - 7.5 kg per box × 7 boxes = 52.5 kg

How do I find which item is the best value?

- A common proportion question is to find which product is the **best value for money**
  - Which is best value for money?
    - 1.5 kg of flour for £0.84
    - or 5 kg of flour for £2.65?
- To do this, compare the prices of 1 kg (1 **unit**):
  - Divide the price in dollars by the weight this buys
    - £0.84 ÷ 1.5 kg = £0.56 per kg
    - £2.65 ÷ 5 kg = £0.53 per kg
  - Make sure you compare prices for the **same size unit** of both items
    - The lowest unit price is the **best** value (0.53 < 0.56)
    - 5 kg of flour for £2.65 is **better value for money** than 1.5 kg of flour for £0.84

Examiner Tips and Tricks

Start a question by jotting down the **key values** involved. You'll be able to pick them out quickly and easily later on.

Worked Example

The bonus received by an employee is directly proportional to the profit made by the company they work for.
Bonuses are paid at a rate of £250 per £3000 profit the company makes.

(i) Work out the bonus an employee receives if the company makes a profit of £18 000.

(ii) If the company makes less than £600 profit, no bonus is paid. Find the lowest bonus an employee could receive.

Answer:

(i) Identify the two quantities 'profit' and 'bonus'

Find the factor ('new' ÷ 'old') from the profit

Multiply the bonus by the factor

$$250 \times 6 = 1500$$

Answer in context with units

An employee should receive a bonus of £1500

(ii) We are still working with profit and bonus

The lowest bonus will be when the company makes exactly £600 profit

Find the factor using 'new' ÷ 'old'

$$\frac{600}{3000} = \frac{1}{5}$$

Find the amount of bonus by multiplying by the factor

$$250 \times \frac{1}{5} = 50$$

Answer in context with units

The lowest amount of bonus an employee could receive is £50

Inverse proportion

What is inverse proportion?

- **Inverse** proportion
  - As one quantity **increases** by a certain rate (factor)
  - The other quantity will **decrease** by the same rate
- This relationship applies vice versa too, if one quantity **decreases** the other **increases**
- E.g. If 2 robots take 15 hours to build a car
  - **Tripling** the number of robots (6) would mean the time taken to build a car would be **divided by 3** (5 hours)

How do I solve inverse proportion questions?

- **Read** through wordy **inverse proportion** questions **carefully**
  - Ensure that you understand the **context** of the question
  - Some questions may tell you the relationship between the two values as a **ratio**
- **Identify** the two quantities involved
- **Find** the **factor** that you will be **increasing/decreasing** by
  - This may be given to you in the question, e.g. 'the amount is tripled'
  - Alternatively, find this by dividing the 'new' quantity by the 'old' quantity
- **Divide** the other quantity by this factor to find the required **quantity**
- Give your final answer in **context**

with inverse proportion?

- This is similar to the unitary method for direct proportion
  - Just remember to do the **opposite** operation when scaling
- For example, 5 workers take 20 hours to complete a job, find how long it would take 8 workers
  - Find the time it would take **1 worker** (1 **unit**)
    - The number of workers decreases
    - Therefore, the time increases
    - 5 × 20 hours = 100 hours
  - Scale this unit up to get the required number
    - The number of workers increases
    - Therefore, the time decreases
    - 100 hours ÷ 8 = 12.5 hours

Examiner Tips and Tricks

Think about the **context** to determine if a question is **direct or inverse proportion**

- As the number of robots goes up, the time to build a car comes down (**inverse proportion**)
- If you buy more boxes of cereal, the amount of cereal also increases (**direct proportion**)

Worked Example

The time taken to fill a swimming pool is inversely proportional to the number of pumps used to pump the water in.
If 3 pumps are used it will take 12 hours to fill the pool.

(i) Work out the amount of time required to fill the pool if 9 pumps are used.

(ii) Work out the minimum number of pumps needed in order to fill the pool in 6 hours.

Answer:

Method 1 (Unitary method)

(i)

Find how long it would take one pump

- The number of pumps decreases by a factor of 3
- The time increases by a factor of 3

$$12 \times 3 = 36$$

1 pump takes 36 hours

Find how long it would take 9 pumps

$$36 \div 9 = 4$$

It takes 4 hours for 9 pumps to fill the swimming pool

(ii)

Find how many pumps are needed to fill the pool in 1 hour

- The time decreases by a factor of 12
- The number of pumps increases by a factor of 12

$$3 \times 12 = 36$$

1 hour requires 36 pumps

Find how many pumps are needed to fill the pool in 6 hours

- The time increases by a factor of 6
- The number of pumps decreases by a factor of 16

$$36 \div 6 = 6$$

6 pumps are needed to fill the pool in 6 hours

Method 2

(i)

Identify the two quantities, 'number of pumps' and 'time (hours)'

Find the factor ('new' ÷ 'old') from the number of pumps

$$\frac{9}{3} = 3$$

Divide the time by the factor

$$12 \div 3 = 4$$

Answer in context with units

It takes 4 hours for 9 pumps to fill the swimming pool

(ii)

We are still working with 'pumps' and 'time'

Find the factor for time using 'new' ÷ 'old'

$$\frac{6}{12} = \frac{1}{2}$$

Find the number of pumps by dividing by the factor

$$3 \div \frac{1}{2} = 6$$

6 pumps are needed to fill the pool in 6 hours

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